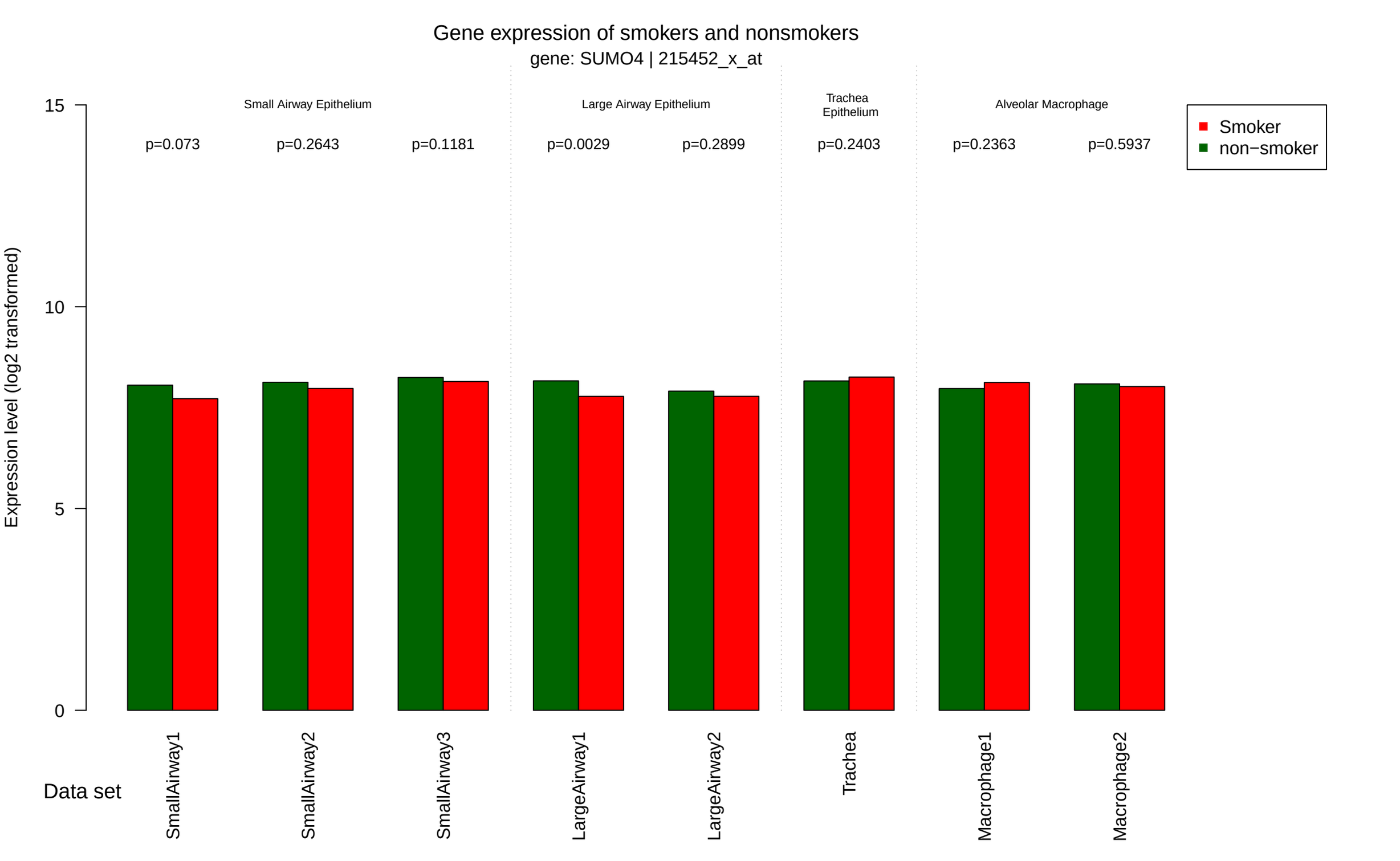
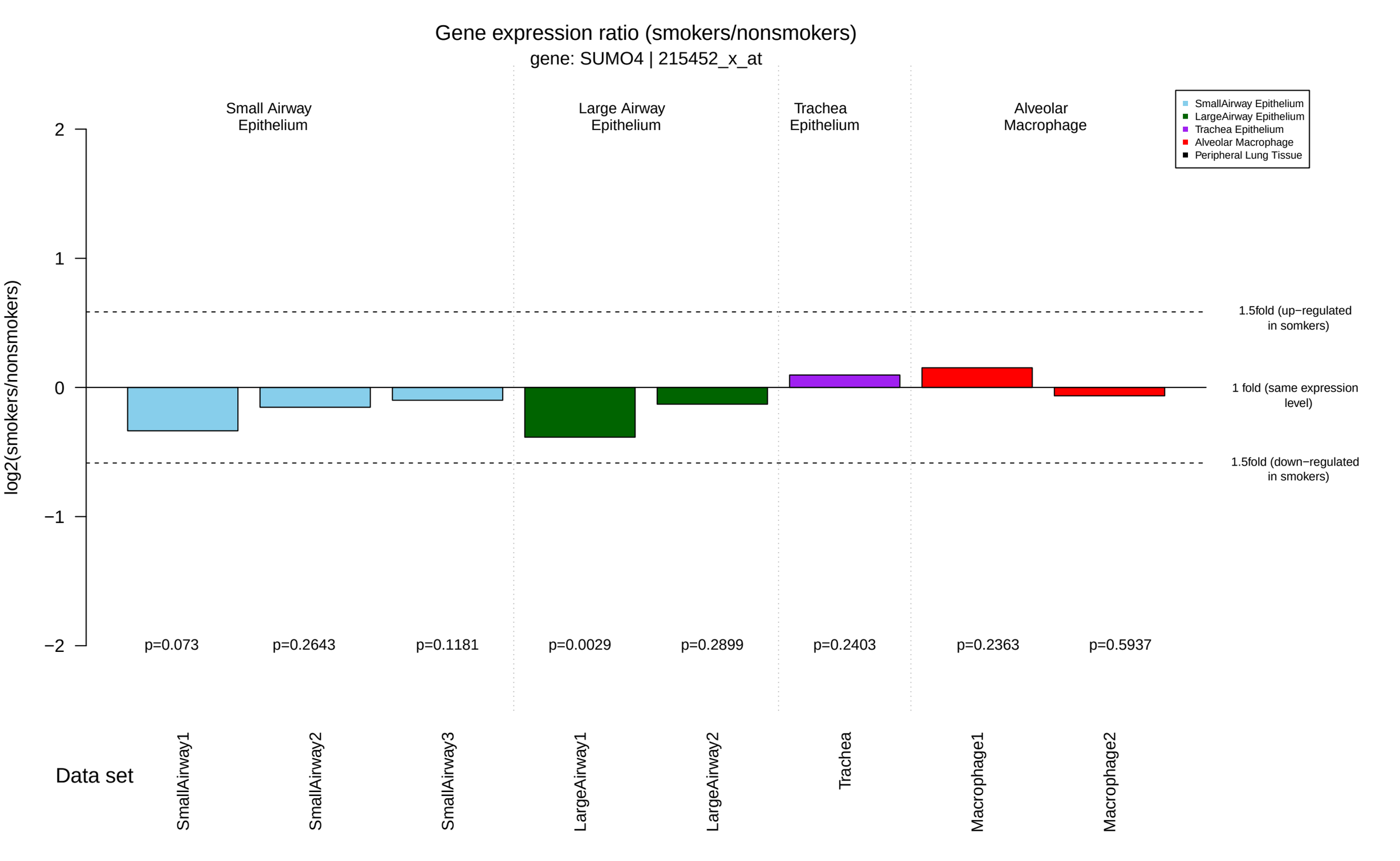
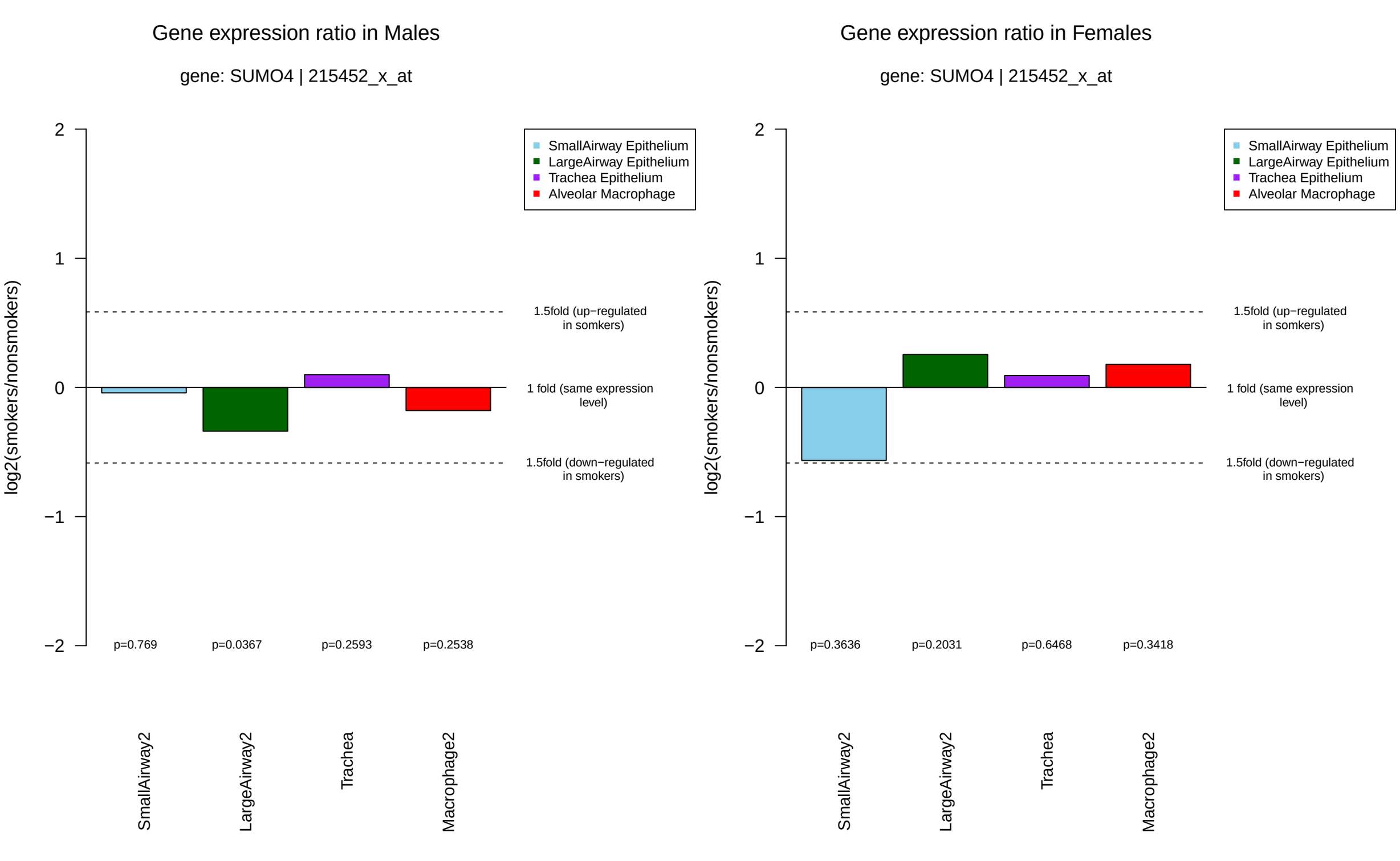
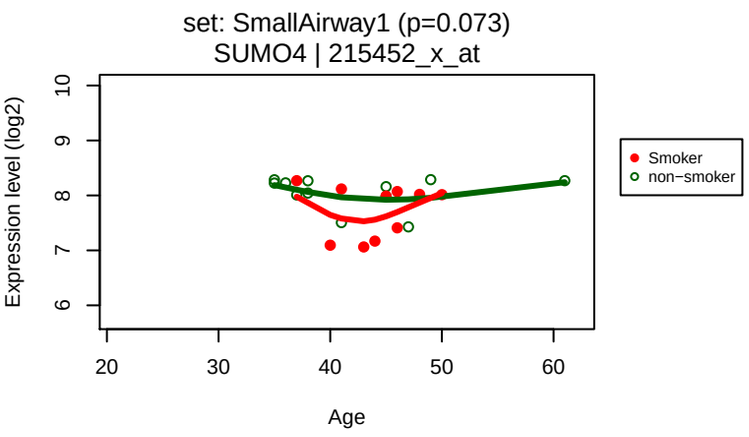
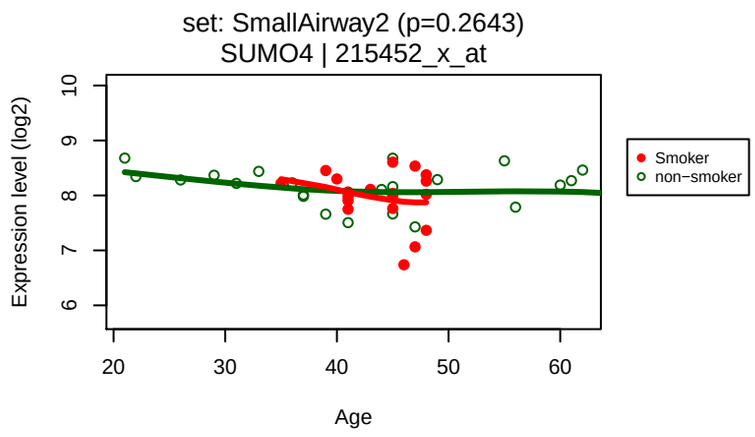
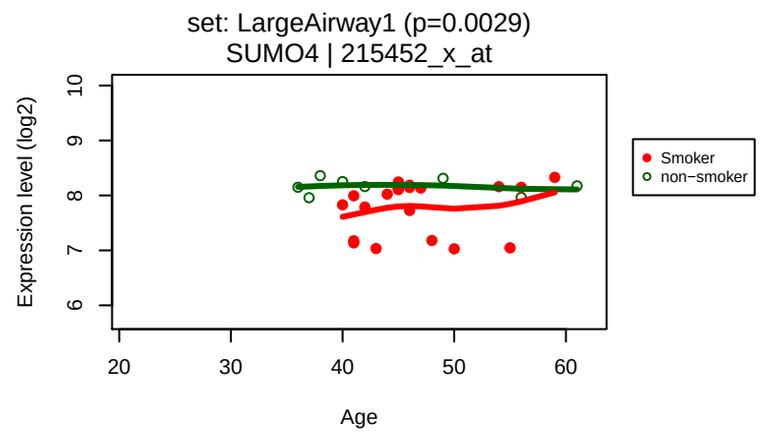

Gene expression of smokers and nonsmokers
gene: SUMO4 | 215452_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.073
p=0.2643
p=0.1181
p=0.0029
p=0.2899
p=0.2403
p=0.2363
p=0.5937
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

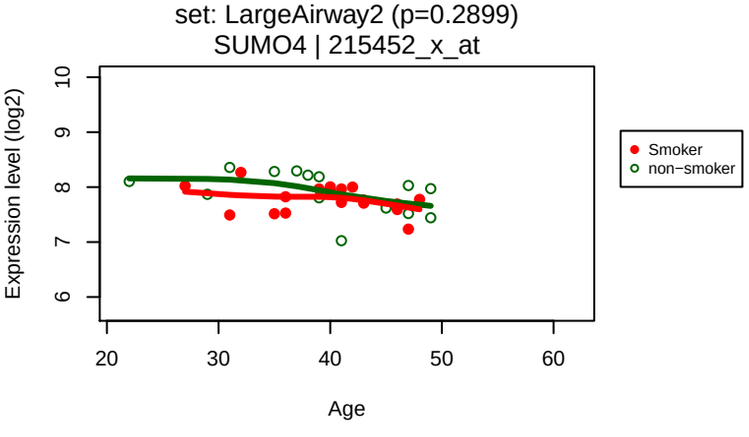

Gene expression ratio (smokers/nonsmokers)
gene: SUMO4 | 215452_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.073
p=0.2643
p=0.1181
p=0.0029
p=0.2899
p=0.2403
p=0.2363
p=0.5937
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

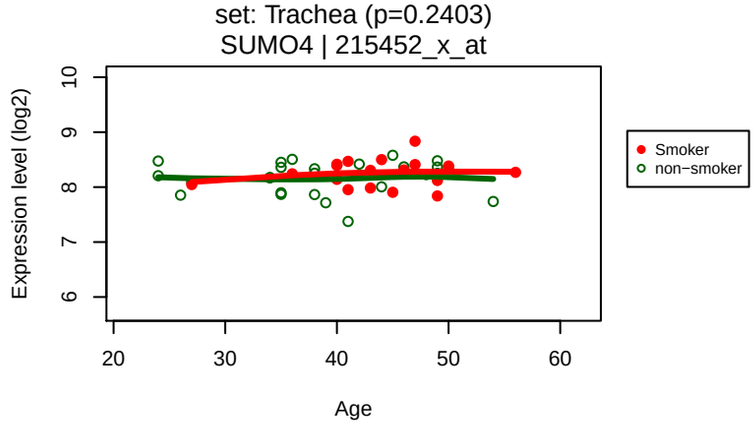

Gene expression ratio in Males
gene: SUMO4 | 215452_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.769
p=0.0367
p=0.2593
p=0.2538
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SUMO4 | 215452_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3636
p=0.2031
p=0.6468
p=0.3418
SmallAirway2
LargeAirway2
Trachea
Macrophage2

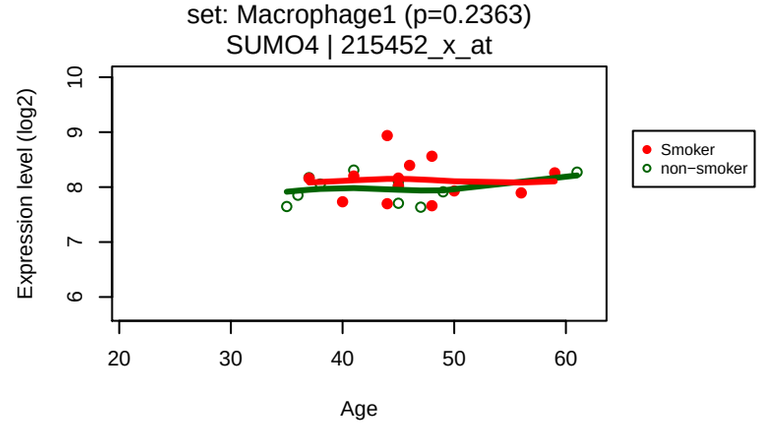

set: SmallAirway1 (p=0.073)
SUMO4 | 215452_x_at
Expression level (log2)
Age
Smoker
non−smoker

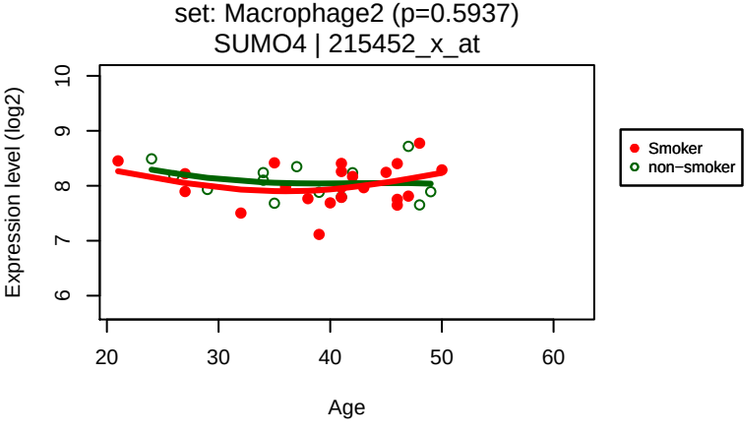

set: SmallAirway2 (p=0.2643)
SUMO4 | 215452_x_at
Expression level (log2)
Age
Smoker
non−smoker

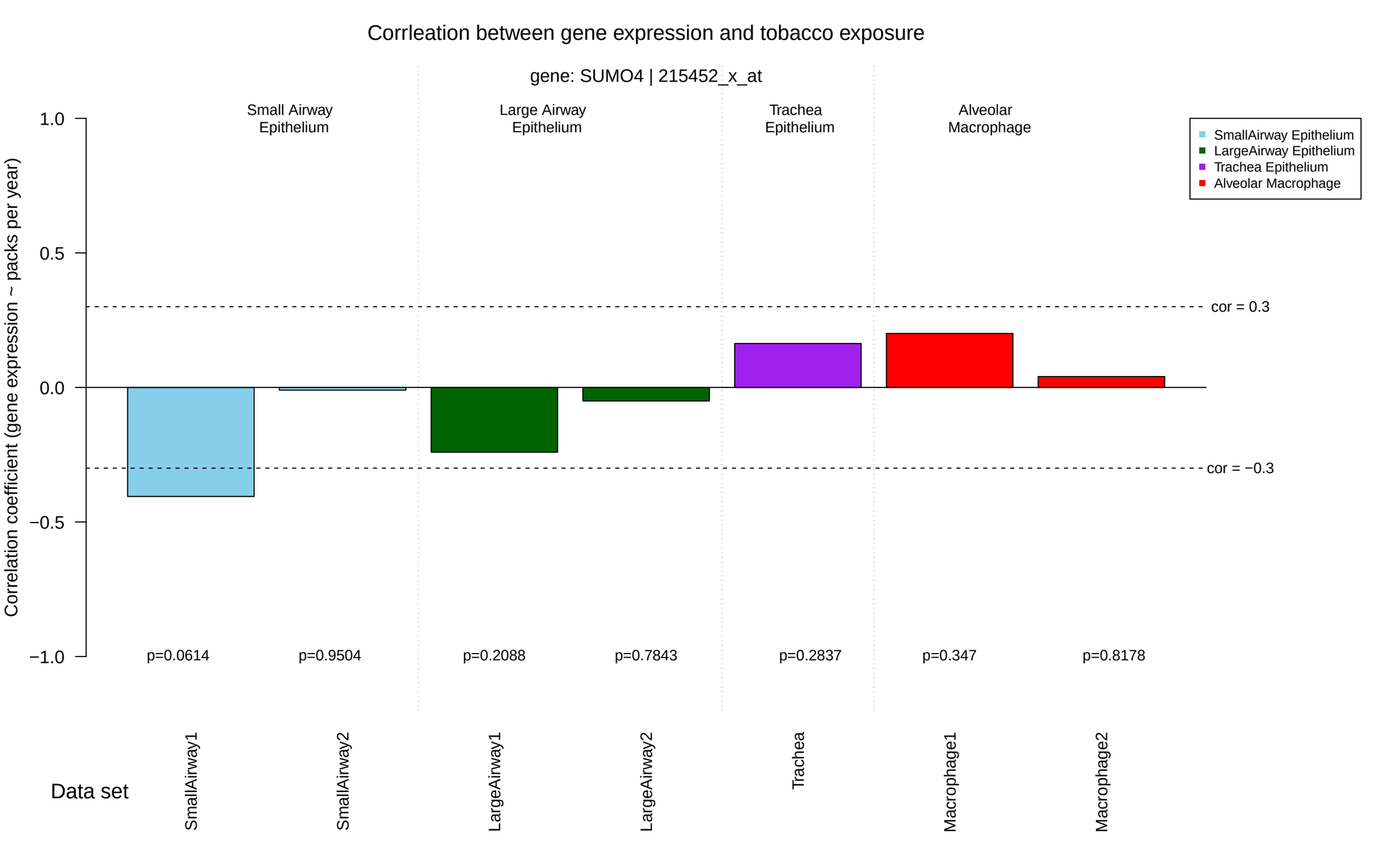

set: LargeAirway1 (p=0.0029)
SUMO4 | 215452_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2899)
SUMO4 | 215452_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2403)
SUMO4 | 215452_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2363)
SUMO4 | 215452_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5937)
SUMO4 | 215452_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SUMO4 | 215452_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0614
p=0.9504
p=0.2088
p=0.7843
p=0.2837
p=0.347
p=0.8178
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2